Teaching Quantum Chemistry to a Deep Learning Model

Transformers for the many body Schrodinger Equation

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Outline

1. The Schrödinger Wave Function and the physical laws that rule
   - Schrödinger Equation
   - Physical laws and conditions
   - Optimizing an Ansatz
2. Transformers
   - Attention Mechanism
3. Psiformer
   - Fermi Net
   - Psi Former
   - Practical Implementation Details

The Schrödinger equation

On 1926 Schrodinger derived his equation: Let Ψ be a complex value function called **wave function**, $\hat{H}$ is called the **Hamiltonian Operator** and E is the energy of the system, then the Schrödinger equation is:

$$\hat{H}\,\Psi(\mathbf{R}) = E\Psi(\mathbf{R}) \tag{1}$$

In this slide we consider the **single-particle** case, so $\mathbf{R} = \mathbf{r} \in \mathbb{R}^3$.

Hamiltonian

$$\hat{H} = \frac{\hat{\vec{P}}^2}{2m} + \hat{V} = -\frac{\hbar^2}{2m}\nabla^2 + \hat{V} \tag{2}$$

Where $\hat{V}$ is the potential operator and ∇ is the Laplacian operator.

Wave Function as Probability Density

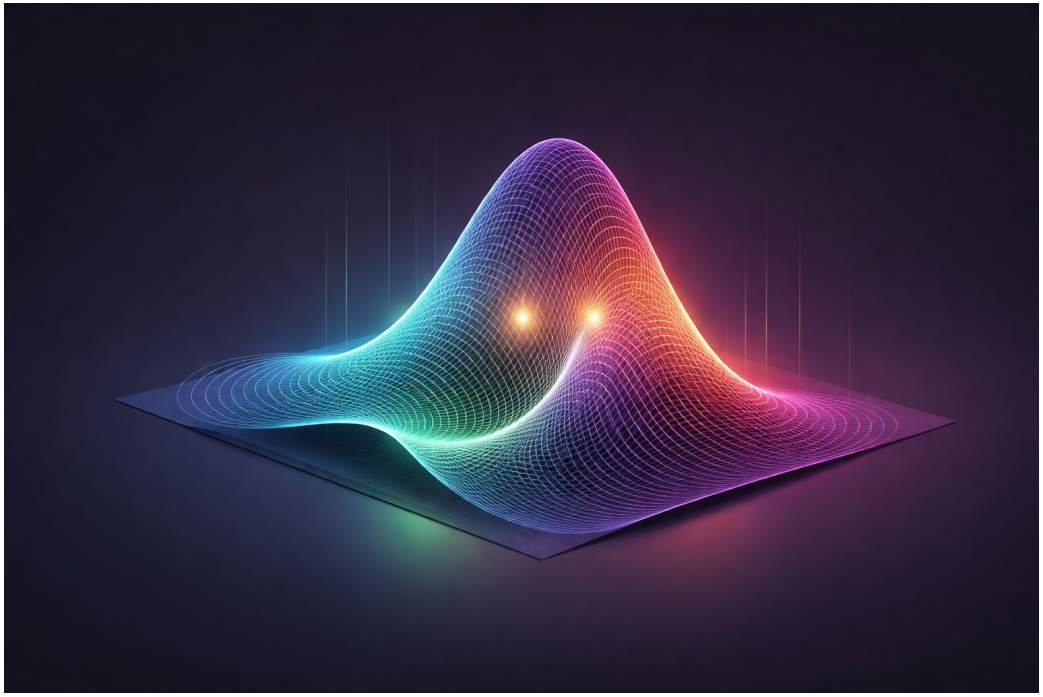

Figure 1: $|\Psi(\mathbf{R})|^2$ represent the probability to find a particle near the position $\mathbf{R}$.

Many-Body System

When considering n bodies, we have:

$$\hat{H}\psi(\mathbf{x}_1, \ldots, \mathbf{x}_n) = E\psi(\mathbf{x}_1, \ldots, \mathbf{x}_n) \tag{3}$$

With $\mathbf{x}_i = \{\mathbf{r}_i, \sigma\}$, where $\mathbf{r}_i \in \mathbb{R}^3$ is the position of each particle and $\sigma \in \{\uparrow . \downarrow\}$ is the so called spin. The Hamiltonian becomes hard because all interaction terms are coupled across particles.

Considerations

- Each particle interacts with all other particles.
- For atoms, we include electron-electron, electron-nucleus, and nucleus-nucleus interactions.
- The solution must obey physical laws.

Setting up the Hamiltonian

Start from the full non-relativistic molecular Hamiltonian (atomic units):

$$\hat{H} = -\sum_{i=1}^{N} \frac{1}{2}\nabla_i^2 - \sum_{I=1}^{N_n} \frac{1}{2M_I}\nabla_I^2 - \sum_{i=1}^{N}\sum_{I=1}^{N_n} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \sum_{1\leq i<j\leq N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \quad 1\leq \tag{4}$$

- The hard part is the potential operator: every electron interacts with all other charges.

Oppenheimer approximation

Assume nuclei are much heavier than electrons and treat nuclear positions as fixed parameters.

$$\hat{H}_e = -\sum_{i=1}^{N} \frac{1}{2}\nabla_i^2 - \sum_{i=1}^{N}\sum_{I=1}^{N_n} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \sum_{1\leq i<j\leq N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + E_{nn}(\mathbf{R}) \quad (5)$$

where

$$E_{nn}(\mathbf{R}) = \sum_{1\leq I<J\leq N_n} \frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|}$$

is constant for fixed nuclei.

Fermi-Dirac statistics

All fermions follow Fermi-Dirac statistics.

Antisymmetry requirement

Exchanging two electrons must flip the sign of the wavefunction:

$$\Psi(\dots, \mathbf{x}_i, \mathbf{x}_j, \dots) = -\Psi(\dots, \mathbf{x}_j, \mathbf{x}_i, \dots)$$

- This is a hard structural constraint on the ansatz.

Slater Determinant

One standard way to enforce antisymmetry is a determinant:

$$\psi = \begin{vmatrix} \phi_1^k(\mathbf{x}_1) & \dots & \phi_1^k(\mathbf{x}_n) \\ \vdots & & \vdots \\ \phi_n^k(\mathbf{x}_1) & \dots & \phi_n^k(\mathbf{x}_n) \end{vmatrix} \tag{6}$$

where ϕ are spin orbitals.

Kato cusp conditions

Near coalescence, Coulomb terms

$$V \sim \frac{1}{r}$$

make the wavefunction non-smooth at short distance.

- Intuition: when two charged particles get very close, the potential changes very fast.
- Therefore, Ψ is continuous but its radial derivative has a finite jump (a **cusp**).

Kato cusp constraints (atomic units)

$$\left.\frac{\partial_r \Psi}{\Psi}\right|_{r_{iI}\to 0} = -Z_I \qquad \left.\frac{\partial_r \Psi}{\Psi}\right|_{r_{ij}\to 0} = \frac{1}{2}$$

Note: for electron-electron pairs, opposite-spin cusp is stronger; same-spin is softened by antisymmetry.

Any good ansatz should satisfy these limits to get accurate local energies.

From Operator to Local Quantity

For any operator $\hat{O}$, define its local form on the wavefunction:

$$O_L(\mathbf{R}) = \frac{\hat{O}\,\Psi_\theta(\mathbf{R})}{\Psi_\theta(\mathbf{R})}$$

Using

$$p_\theta(\mathbf{R}) = |\Psi_\theta(\mathbf{R})|^2 \frac{1}{\int d\mathbf{R}'\,|\Psi_\theta(\mathbf{R}')|^2},$$

the expectation value can be estimated as

$$\langle \hat{O} \rangle_\theta = \mathbb{E}_{\mathbf{R}\sim p_\theta}[O_L(\mathbf{R})]\,.$$

Loss: Variational Principle

Choose the specific operator $\hat{O} = \hat{H}$ (Hamiltonian), so

$$E_L(\mathbf{R}) = \frac{\hat{H}\,\Psi_\theta(\mathbf{R})}{\Psi_\theta(\mathbf{R})}.$$

Then the variational objective is

$$E[\Psi_\theta] = \frac{\langle \Psi_\theta | \hat{H} | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} = \mathbb{E}_{\mathbf{R}\sim p_\theta}[E_L(\mathbf{R})] \geq E_0.$$

Minimizing this energy drives the ansatz toward the ground state.

Variational Monte Carlo

Quantum Monte Carlo

With the samples $\mathbf{R}_1, \ldots, \mathbf{R}_M \sim |\Psi|^2_\theta(\mathbf{R})$ we can make:

$$\mathcal{L}_\theta = \mathbb{E}_{\mathbf{R}\sim\Psi^2_\theta}[E_L(\mathbf{R})] \approx \frac{1}{M}\sum_{i=1}^{M} E_L(\mathbf{R}_i) \tag{7}$$

With:

$$E_L(\mathbf{R}_k) = \frac{\hat{H}\psi(\mathbf{R}_k)}{\psi(\mathbf{R}_k)} = -\frac{1}{2}\frac{\nabla^2\psi(\mathbf{R_k})}{\psi(\mathbf{R}_k)} + V(\mathbf{R}_k)$$

Obtain $\mathbf{R}_k \rightarrow$ Metropolis-Hastings Algorithm

Metropolis-Hastings Algorithm

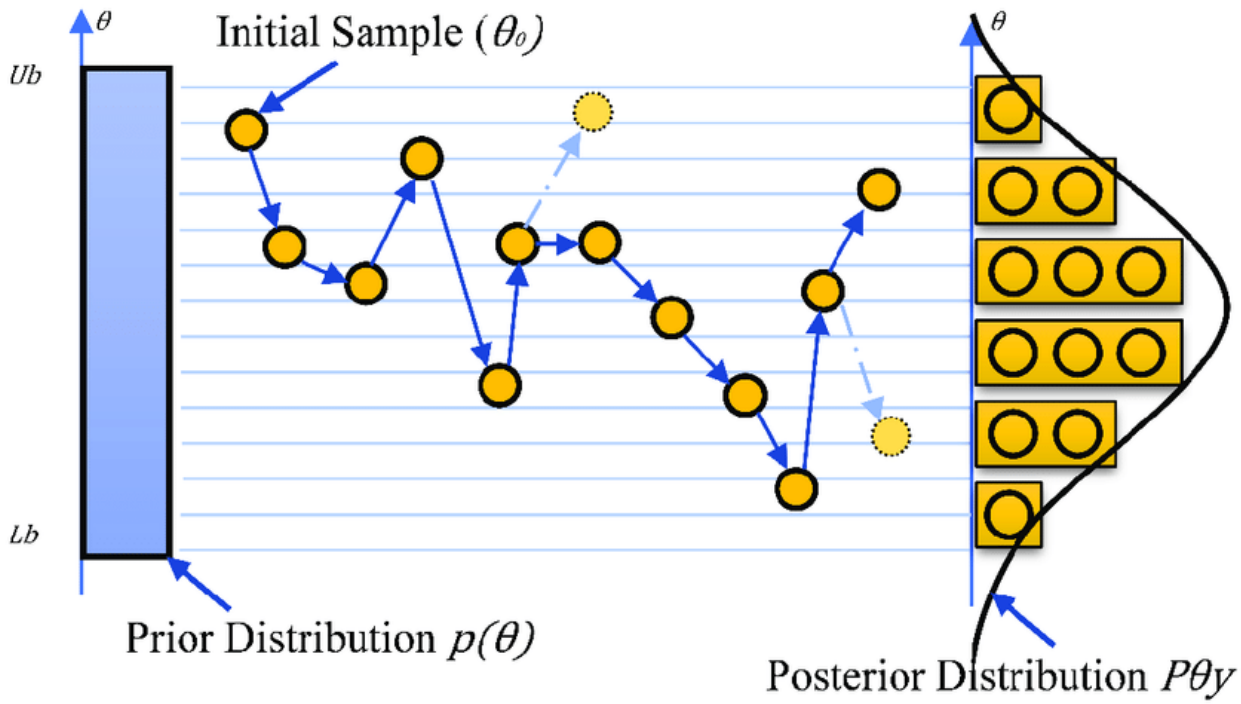

Figure 2: Metropolis-Hastings Walkers

- Obtain its own data (no dataset $\mathcal{D}$), $\rightarrow \mathcal{L}_\theta = \mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}[E_L(\mathbf{R})]$.
- Third order derivatives
- $|\Psi_\theta|^2$ changes over time, you are just optimizing just the **energy**, not the wave function itself.

Metropolis-Hastings Algorithm

Goal: Generate many samples $\mathbf{R} \sim \rho$, Requirement: $C\rho$
1. $\mathbf{R}_0 \in E$ arbitrary: 2. Propose $\mathbf{R}' = \mathbf{R}_0 + \eta$,where $\eta \sim q(\eta)$, (Normal Gaussian) 3. Compute the quantity:

$$A(\mathbf{R_0}, \mathbf{R}') = \min\left(1, \frac{\rho(\mathbf{R}')}{\rho(\mathbf{R}_0)}\right)$$

4. Generate a uniform number $U \in [0, 1]$. If: $U < A(\mathbf{R}_0, \mathbf{R}')$ then $\mathbf{R_1} = \mathbf{R}'$, otherwise try another $\mathbf{R}'$. Accept or decline.

Solution: Log Derivative Trick

$$\mathcal{L}_\theta = \mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}[E_L(\mathbf{R})]$$

$$\downarrow$$

$$\nabla_\theta \mathcal{L} = 2\mathbb{E}_{\mathbf{R}\sim\Psi^2}[(E_L(\mathbf{R}) - \mathbb{E}_p[E_L])\nabla_\theta \log\psi]$$

$$\uparrow$$

$$\mathcal{L}(\theta) = 2\mathbb{E}_{\mathbf{R}\sim\Psi^2}[\underbrace{(E_L(\mathbf{R}) - \mathbb{E}_p[E_L])}_{\text{detach}} \log\psi]$$

REINFORCE

$$J(\theta) = \mathbb{E}_\tau[R(\tau)]$$

$$\downarrow$$

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim\pi}[(R(\tau) - b)\nabla_\theta \log\pi_\theta(\tau)]$$

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Transformer Architecture

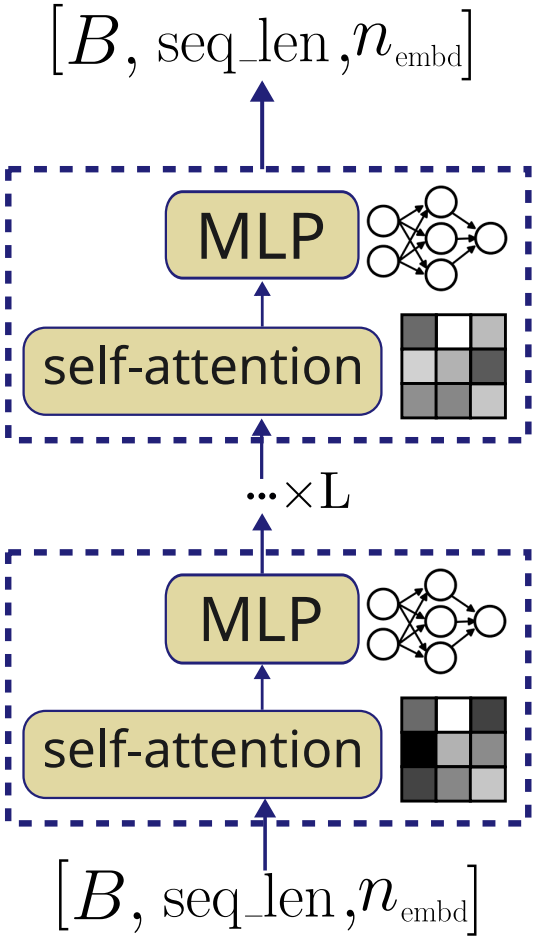

Figure 3: Tranformer backbone

Multi Head Attention → Self Attention

- n_{embd} the embedding dimension
- n_h the number of attention heads
- d_h the dimension per head
- $\mathbf{h}_t \in \mathbb{R}^{n_{\text{embd}}}$ the hidden dimension.

Attention on the room

The learnable matrices are:

$$W^Q, W^K, W^V \in \mathbb{R}^{n_{\text{embd}} \times n_{\text{embd}}}$$

$$\mathbf{k}_i = \mathbf{W}^k \mathbf{h}_i, \mathbf{q}_i = \mathbf{W}^q \mathbf{h}_i, \mathbf{v}_i = \mathbf{W}^v \mathbf{h}_i$$

$$[\mathbf{q_1}, \mathbf{q_2}, \ldots, \mathbf{q_{n_h}}] = \mathbf{q}$$
$$[\mathbf{k_1}, \mathbf{k_2}, \ldots, \mathbf{k_{n_h}}] = \mathbf{k}$$
$$[\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_{n_h}}] = \mathbf{v}$$

In the $i - th$ head:

$$\mathbf{o}_{t,i} = \sum_{j=1}^{t} \text{Softmax}\left(\frac{\mathbf{q}_{t,i}^T \mathbf{k}_{j,i}}{\sqrt{d_h}}\right) \mathbf{v}_{j,i} \tag{8}$$

W^O the output projection matrix.

$$\mathbf{u}_t = W^O [\mathbf{o}_{t,1}; \mathbf{o}_{t,2}; \ldots; \mathbf{o}_{t,n_h}]$$

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Psiformer Ansatz

Ansatz: Proposal model that you propose guided by intuition and that you optimize.

- be *antisymmetric* under particle exchange,
- capture strong *electron–electron correlations*.

Proposed ansatz

Motivated by these constraints, we propose a Slater–Jastrow form:

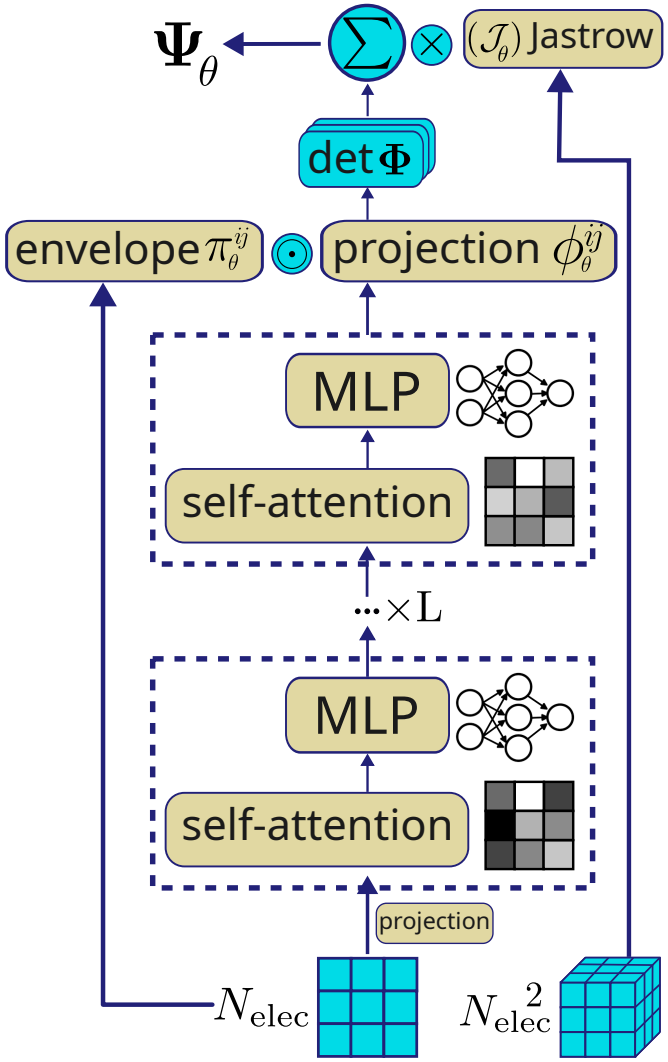
$$\Psi_\theta(\mathbf{R}) = \underbrace{\exp\left(\mathcal{J}_\theta(\mathbf{R})\right)}_{\text{Coulomb correlations}} \times \underbrace{\sum \omega_k \det\left[\boldsymbol{\phi}_\theta^k(\mathbf{R})\right]}_{\text{antisymmetry}}$$

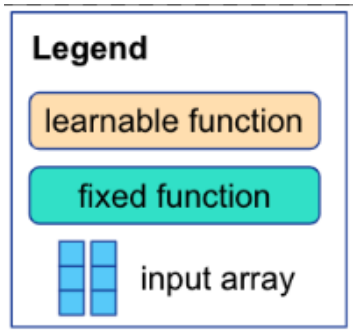
Jastrow Factor $\exp(\mathcal{J})$

In this work we are going to use this specific form:

$$\mathcal{J}_\theta(x) = \sum_{i<j;\sigma_i=\sigma_j} -\frac{1}{4}\frac{\alpha_{par}^2}{\alpha_{par} + |\mathbf{r}_i - \mathbf{r}_j|} + \sum_{i,j;\sigma_i\neq\sigma_j} -\frac{1}{2}\frac{\alpha_{anti}^2}{\alpha_{anti} + |\mathbf{r}_i - \mathbf{r}_j|} \quad (9)$$

Psi Former Architecture





- The envelope is a matrix with learned exponential decay as elements.
- Learn the Jastrow factor only have two parameters.
- In practice we use logarithm scale.

Figure 4: Psi Former Architecture

Psiformer Shapes

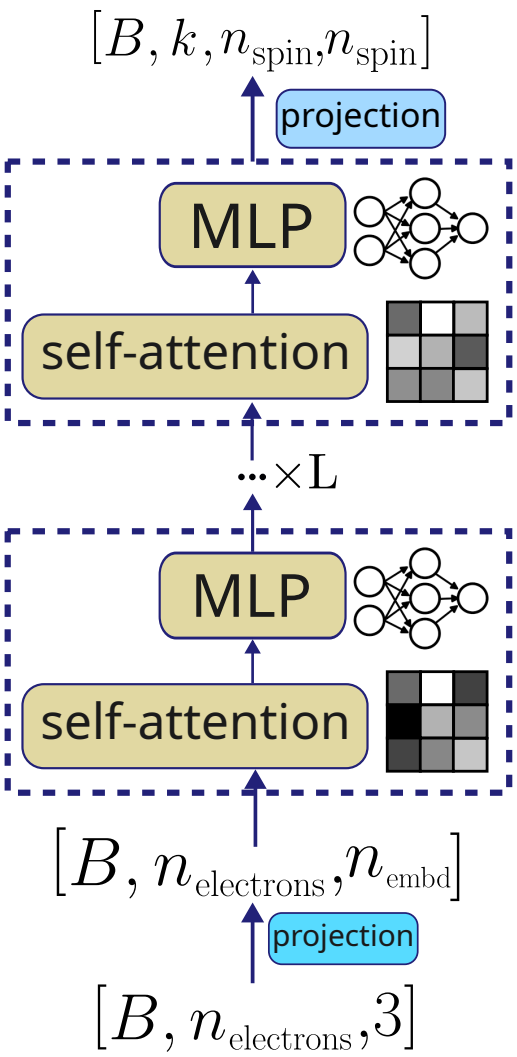


Figure 5: Psiformer shapes handling

Methodology

Hyperparameter	Small	Large
Layers L	2	4
Heads H	4	8
Model Dim d	256	512
MLP Dim d_{ff}	1024	2048
Determinants K	1	2
MCMC walkers N_w	1024	2048
MCMC steps / iter	10	10
Learning rate	2×10^{-4}	1×10^{-4}
Total parameters	50441	$3M$

Figure 6: Psiformer Torch Small and Large

Goal: Obtain accurate ground state-energies using an Ansatz created with **Torch** library

Training Loop:

Set the spins and electron number

1. Sample configurations $\mathbf{R}_k$
2. Estimate local energy E_L
3. Backpropagation using **A.D**
4. Update parameters θ using **AdamGrad**.
5. Track metrics with **Wandb.**

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Implementation : Laplacian Computation

The kinetic energy requires the Laplacian: $\nabla^2\Psi(\mathbf{R}) = \sum_i \frac{\partial^2\Psi}{\partial R_i^2}$

```
R = R_o.requires_grad_(True)      # particle coordinates
psi = model(R)                  # neural wavefunction
# first derivative: gradient
grad_psi = torch.autograd.grad(
    psi, R,
    create_graph=True,
    # retain_graph=True
)[0]
# second derivative: Laplacian
laplacian = 0.0
for i in range(R.shape[-1]):
    laplacian += torch.autograd.grad(
        grad_psi[..., i], R,
        # create_graph=True,
        retain_graph=True
    )[0][..., i]
```

Determinant Stability in Psiformer

$$\Psi_\theta(\mathbf{R}) \propto \det[\mathbf{\Phi}_\theta(\mathbf{R})]$$

Derivative of a determinant.

$$\frac{\partial \det(\mathbf{A})}{\partial \mathbf{A}} = \det(\mathbf{A})\mathbf{A}^{-T}$$

Key instability. If $\mathbf{\Phi}$ becomes singular or nearly singular:

$$\det \mathbf{A} \to 0 \quad \Rightarrow \quad [a_{ii}^{-1}] \to \infty$$

so the gradient can explode even when the wavefunction itself is small.

Fix: Custom Operation

Idea: Use SVD ($A = U\Sigma V^T$) to compute $\nabla \log\det A = A^{-T}$ without explicit inversion (Appendix D).

```
import torch

class StableLogDet(torch.autograd.Function):
    @staticmethod
    def forward(ctx, A):
        # Decompose A: S are singular values
        U, S, Vh = torch.linalg.svd(A)
        ctx.save_for_backward(U, S, Vh)
        return S.log().sum()

    @staticmethod
    def backward(ctx, g):
        U, S, Vh = ctx.saved_tensors
        # Reconstruct A^{-T} = U * diag(1/S) * Vh
        inv_S = torch.diag_embed(1.0 / S)
        grad_A = U @ inv_S @ Vh
        return g * grad_A
```

Optimizer: Adam vs. AdamW

Core Difference: AdamW *decouples* weight decay from the gradient update to fix regularization on adaptive optimizers.

1. **Adam (Entangled L2 Regularization)**
   - Decay is added to the gradient, so it gets scaled by the adaptive variance.

$$g_t = \nabla \mathcal{L} + \lambda \theta_t$$
$$\theta_{t+1} = \theta_t - \text{AdamStep}(g_t)$$

2. **AdamW (Decoupled Weight Decay)**
   - Decay is applied directly, bypassing the adaptive scaling mechanism.

$$g_t = \nabla \mathcal{L}$$
$$\theta_{t+1} = \theta_t - \text{AdamStep}(g_t) - \eta \lambda \theta_t$$

Keeping the GPU Busy: Batched Energy Evaluation

Naive training issue. Initial training evaluated energies step-by-step over MCMC samples.

$$\text{GPU utilization} \approx 30\%$$

Solution: batched evaluation. MCMC samples are reshaped and flattened:

$$(\text{mc_steps}, B, n_e, 3) \;\rightarrow\; (\text{mc_steps} \times B, n_e, 3)$$

Result.

$$\text{GPU utilization} \approx 99\%$$

Results: Convergence Curve

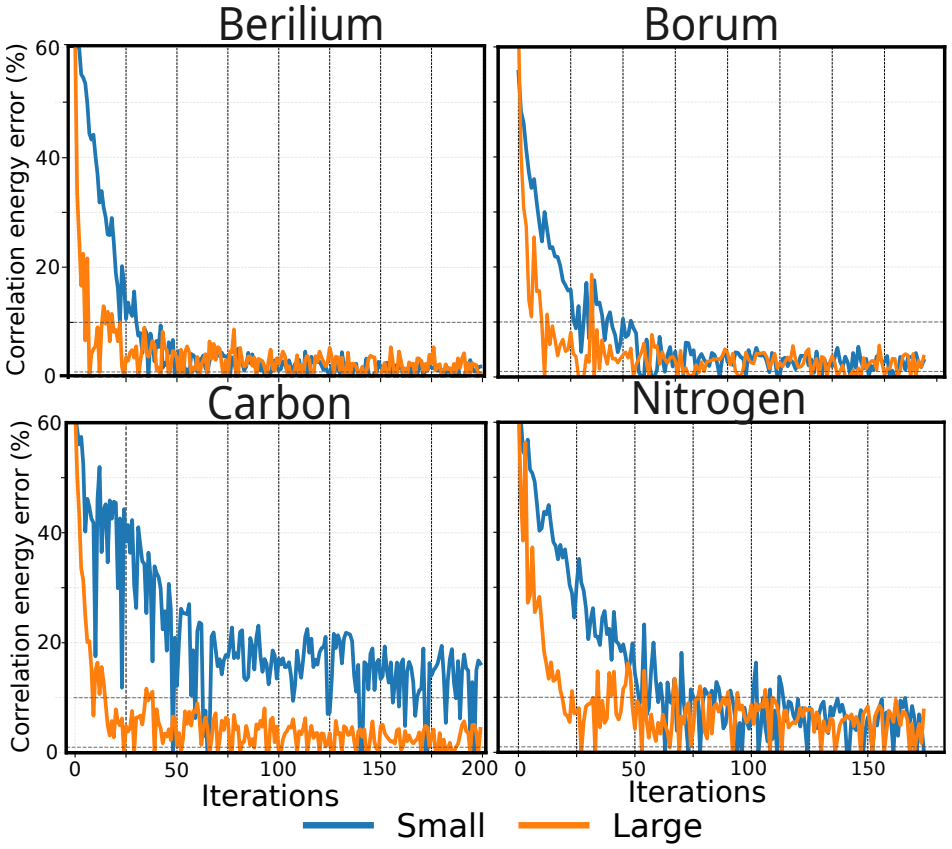


Figure 7: Convergence Curve

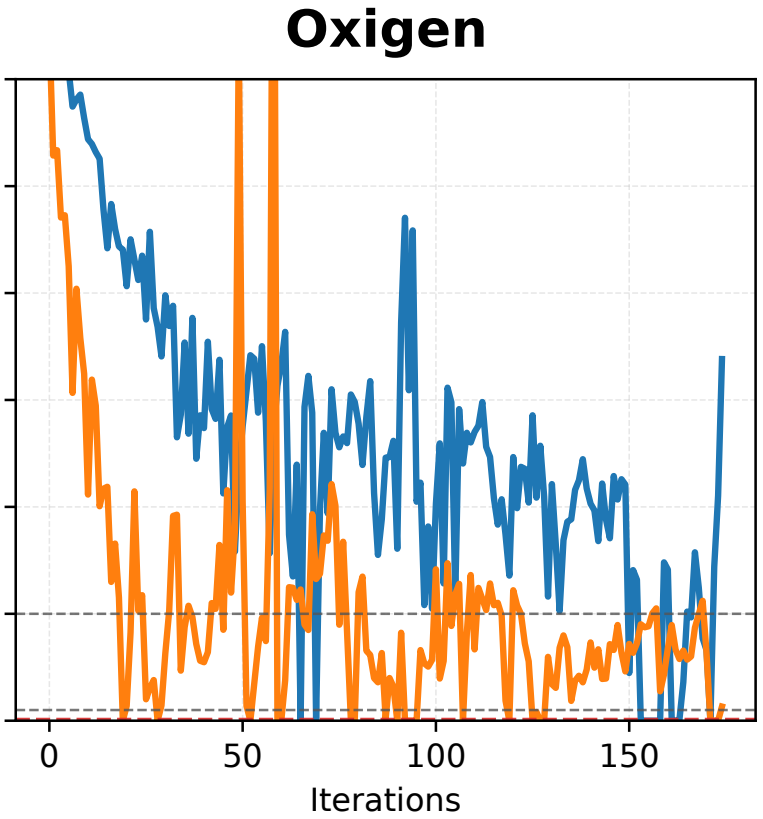


Figure 8: Oxigen Convergence

Results: Energy Estimates

Atom	E_b	E_s	E_l	Δ_s	Δ_l	Δ_{l-s}
H	-0.500	-0.492	**-0.498**	0.008	0.002	**-0.006**
He	-2.903	-2.801	**-2.893**	0.102	0.010	**-0.092**
Li	-7.478	-7.097	**-7.243**	0.381	0.235	**-0.146**
Be	-14.667	-13.901	**-14.237**	0.766	0.430	**-0.336**
B	-24.653	-24.042	**-24.567**	0.611	0.086	**-0.525**
C	-37.845	-35.492	**-36.457**	2.353	1.388	**-0.965**
N	-54.589	-50.492	**-51.700**	4.097	2.889	**-1.208**
O	-75.067	-63.492	**-72.139**	11.575	2.928	**-8.647**

Figure 9: Ground state energies (Ha) for H-O, baseline vs Psiformer

Next Steps

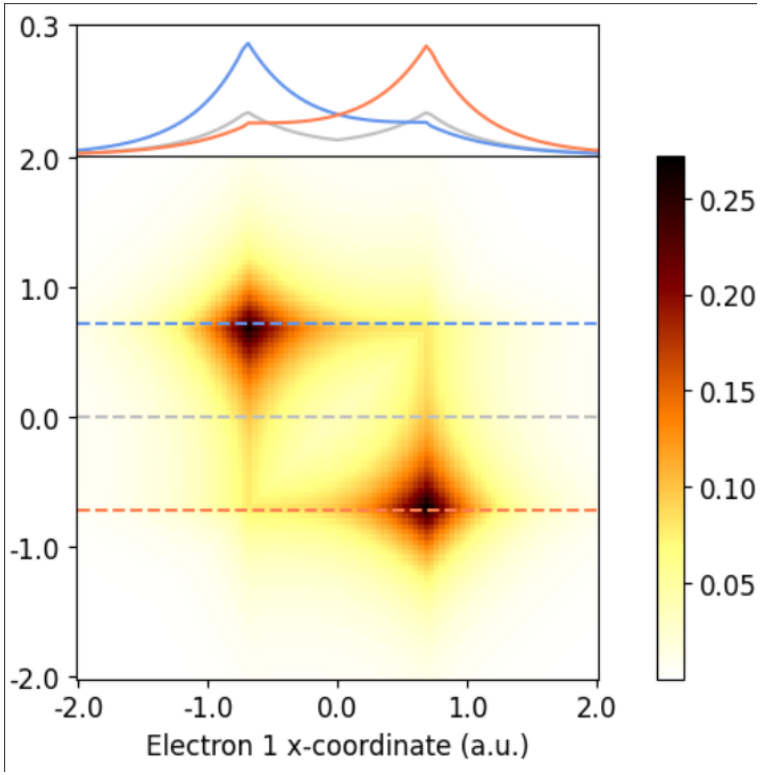
- Build a hybrid framework that combines Density Functional Theory (DFT) with neural wave functions.
- Use DFT orbitals/densities as priors or pretraining targets for faster convergence.
- Reduce the Laplacian bottleneck (e.g., efficient second-order derivatives).
- Explore better optimizers (natural-gradient variants such as KFAC/KFCA).
- Improve scaling with larger systems and transferable training.

References

F. Hermann, Z. Schätzle, and F. Noé,
A Self Attention Ansatz for Ab Initio Quantum Chemistry,
arXiv:2309.12345, 2023.

J. Pfau, J. S. Spencer, A. G. D. G. Matthews, and W. M. C. Foulkes,
Ab Initio Solution of the Many-Electron Schrödinger Equation with Deep Neural Networks,
Physical Review Research, 2, 033429, 2020.

A. Vaswani et al.,
Attention Is All You Need,
Advances in Neural Information Processing Systems (NeurIPS), 2017.

Thanks!

I thank the Computer Science Faculty for providing access to GPU resources, in particular NVIDIA GeForce RTX 4080 Super which enabled the training and evaluation of Psiformer models reported in this work.